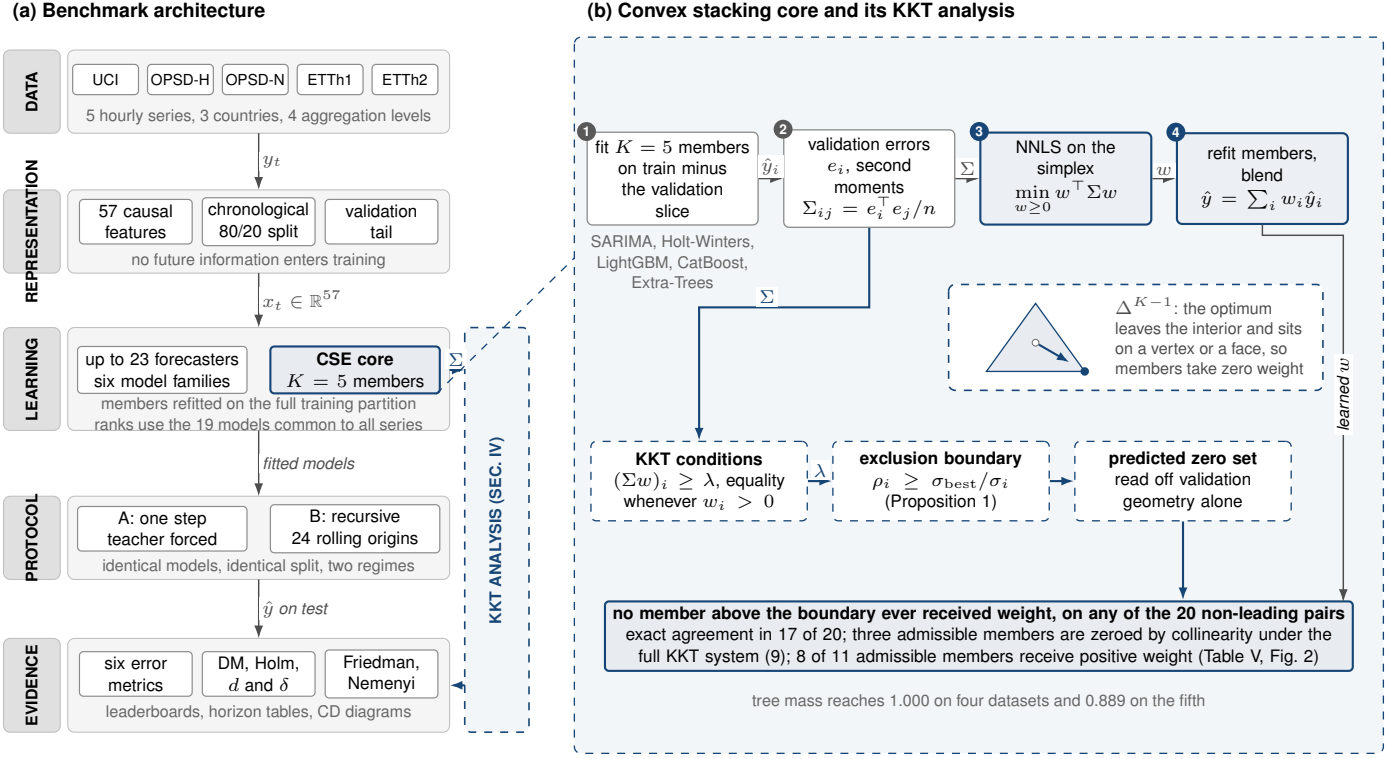

(a) Benchmark architecture
DATA
UCI
OPSD-H
OPSD-N
ETTh1
ETTh2
5 hourly series, 3 countries, 4 aggregation levels
y_t
REPRESENTATION
57 causal features
chronological 80/20 split
validation tail
no future information enters training
$x_t \in \mathbb{R}^{57}$
LEARNING
up to 23 forecasters six model families
CSE core $K = 5$ members
members refitted on the full training partition
ranks use the 19 models common to all series
fitted models
PROTOCOL
A: one step teacher forced
B: recursive 24 rolling origins
identical models, identical split, two regimes
$\hat{y}$ on test
EVIDENCE
six error metrics
DM, Holm, d and δ
Friedman, Nemenyi
leaderboards, horizon tables, CD diagrams
Σ
KKT ANALYSIS (SEC. IV)
(b) Convex stacking core and its KKT analysis
1 fit $K = 5$ members on train minus the validation slice
SARIMA, Holt-Winters, LightGBM, CatBoost, Extra-Trees
$\hat{y}_i$
2 validation errors e_i, second moments $\Sigma_{ij} = e_i^\top e_j / n$
Σ
3 NNLS on the simplex $\min_{w \geq 0} w^\top \Sigma w$
w
4 refit members, blend $\hat{y} = \sum_i w_i \hat{y}_i$
Σ
Δ^{K-1}: the optimum leaves the interior and sits on a vertex or a face, so members take zero weight
learned w
KKT conditions $(\Sigma w)_i \geq \lambda$, equality whenever $w_i > 0$
λ
exclusion boundary $\rho_i \geq \sigma_{\text{best}} / \sigma_i$ (Proposition 1)
predicted zero set read off validation geometry alone
no member above the boundary ever received weight, on any of the 20 non-leading pairs
exact agreement in 17 of 20; three admissible members are zeroed by collinearity under the full KKT system (9); 8 of 11 admissible members receive positive weight (Table V, Fig. 2)
tree mass reaches 1.000 on four datasets and 0.889 on the fifth